

Coding & Solution

Date	03 October 2026
Team ID	COMICCRAFT-TEAM
Project Name	ComicCrafter AI – AI-Powered Comic Strip Generator
Maximum Marks	3 Marks

Core Implementation

1. Streamlit Application – app.py

Provides the user interface, collects the story prompt, offers the four art styles, invokes the backend modules, displays the final 3×2 comic strip and exposes the PDF download button.

2. Gemini Panel Generation – BACKEND/generate_panels.py

Builds a structured prompt requiring exactly six panels, each with a Description and Text field. It calls Gemini 1.5 Flash and uses regular expressions to parse the generated response. A ValueError is raised if the result is not exactly six panels.

3. Image Generation – BACKEND/generate_image.py

Maps the selected art style to a visual description and sends each panel description to the ClipDrop text-to-image endpoint. Generated images are saved under PANEL_IMAGES/.

4. Comic Processing – BACKEND/process_comic.py

Uses Pillow to wrap dialogue, add a text area below each image and combine six processed panels into a 3×2 grid. The final PNG is written to OUTPUT/.

5. PDF Output – app.py

Uses ReportLab SimpleDocTemplate and an A4 page to package the final comic strip as comic_strip.pdf, which is provided through Streamlit's download button.

6. Configuration

python-dotenv loads API keys from the environment. The supplied requirements.txt includes Streamlit, Pillow, requests, python-dotenv, google-generativeai and reportlab.